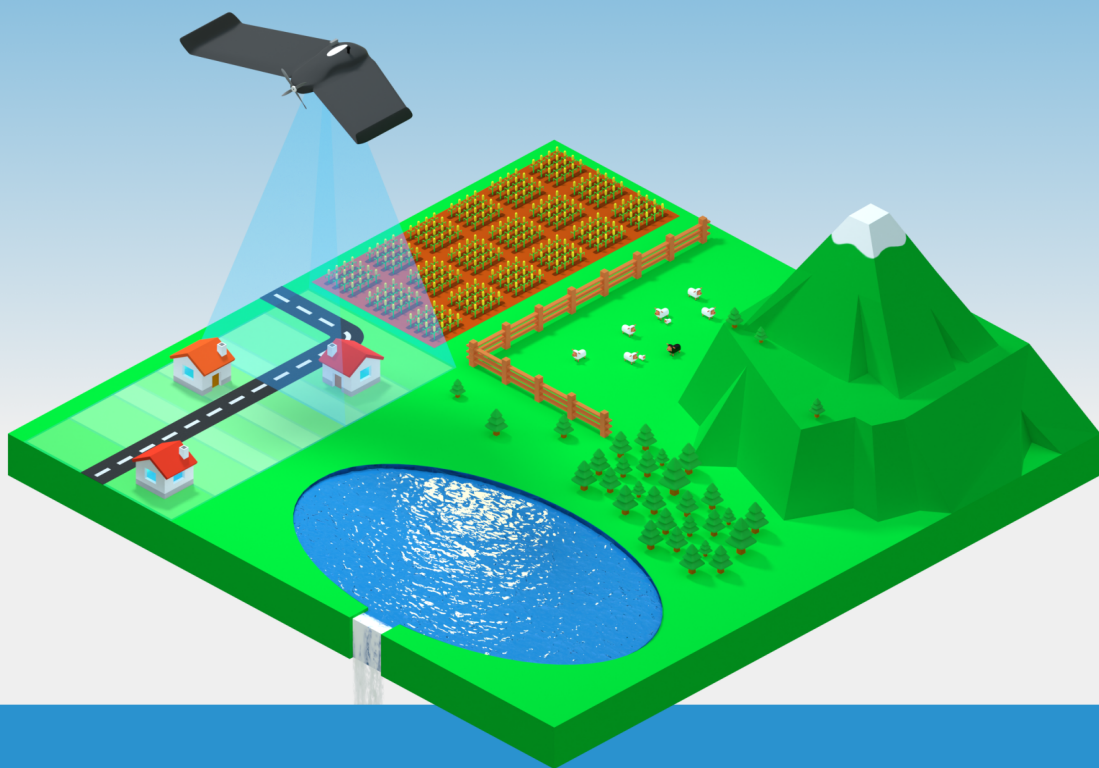


**A Practical Guide to Drone Mapping
Using Free and Open Source Software**



WebODM

The Missing Guide

Third Edition

Piero Toffanin

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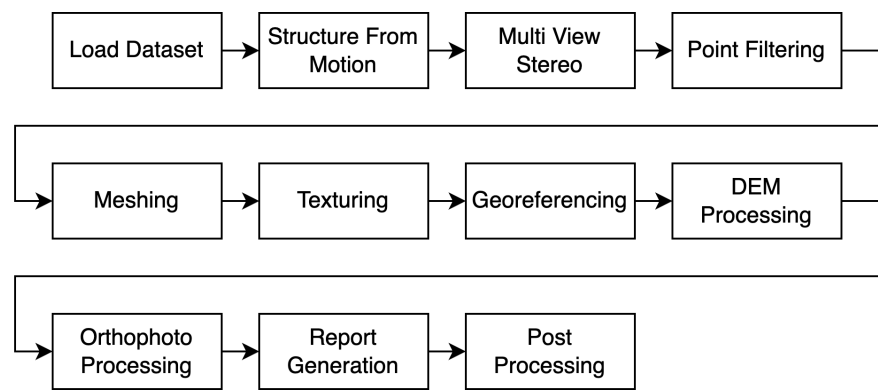
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5. The Processing Pipeline

If this was the first time you’ve used a software like WebODM, or if you can remember the first time you used it, it’s likely that at some point you had a *woah!* moment: how could a computer take simple 2D images and turn them into georeferenced mosaics, 3D models and point clouds?

“Any sufficiently advanced technology is indistinguishable from magic.” -
Arthur C. Clarke

Going from images to 3D models and orthophotos is a process best visualized as a series of incremental steps. Each step relies on the work of previous steps.



WebODM’s processing pipeline

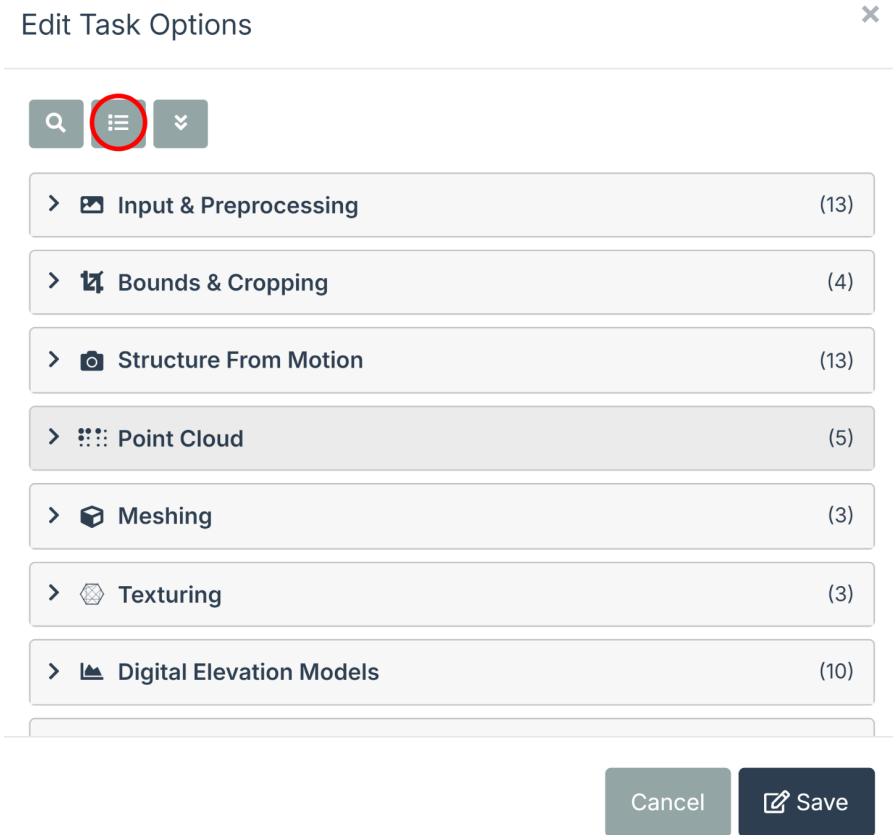
In this chapter we will explore an overview of the pipeline. We will not cover

6. Task Options in Depth

There are several steps involved in the data processing pipeline. Each step has several adjustable settings that influence the output. The software exposes a subset of these available knobs through various options. When creating a task, a user can choose to tweak one or more options to change the behavior of the pipeline.

Options by default are displayed by category, but they can also be displayed alphabetically. To do that, click the list view button from the top right side of the dialog.

6. Task Options in Depth



Options as shown in WebODM when creating a task

Tuning options is more art than science. That's mostly because the best options for certain datasets do not automatically transfer to others. As a general rule, one should start with the defaults, which work fairly well for most datasets and apply tweaks as needed.

A quick note about large language models (LLMs) is warranted here. Many people are turning to these tools for help in picking good processing settings. I

6. Task Options in Depth

must warn that as of 2026 LLMs are *pretty bad* at giving advice when it comes to tuning WebODM options, while sounding very confident in doing so¹.

This chapter is about understanding in detail what each option does. By the end of the chapter you'll be able to quickly improve your results, explain why certain models turn out the way they do and know what to tweak if the results don't turn out the way you want.

A few of these options might be hidden in WebODM and available only from ODX. This is because sometimes the option does not make sense in the context of the graphic interface workflow, or it's simply not supported.

When there is some math to explain, I write the formulas using Python because it's easier than math notation and can be typed on a computer. You can copy/paste the code on a website such as online-python.com and follow along even if you don't know Python.

Feel free to jump around and use this chapter as a reference. As the software gets better, some of these options might disappear from future versions and new ones might be introduced. The list below is taken from the software as of April 28th 2026. In alphabetical order:

3d-tiles

OGC 3D Tiles² are a format specification for visualization and interaction with 3D geospatial content. These files can be displayed with software such as the open source virtual globe engine Cesium³. ODX has support for generating point clouds and textured 3D models in 3D Tiles format by turning on this option (and doesn't rely on third party services to do so).

¹The bs machines: thebullshitmachines.com

²OGC 3D Tiles: ogc.org/standard/3dtiles/

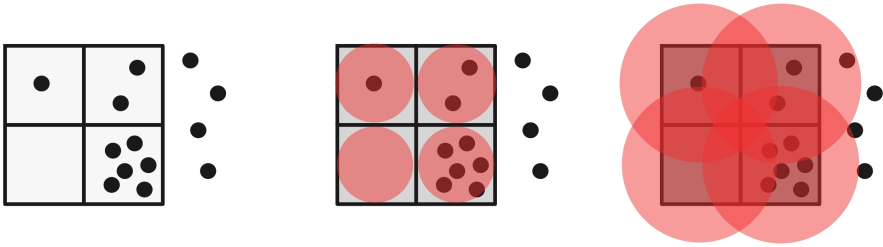
³Cesium: github.com/CesiumGS/cesium

6. Task Options in Depth

Euclidean map results are stored in the *odm_dem* directory.

dem-gapfill-steps

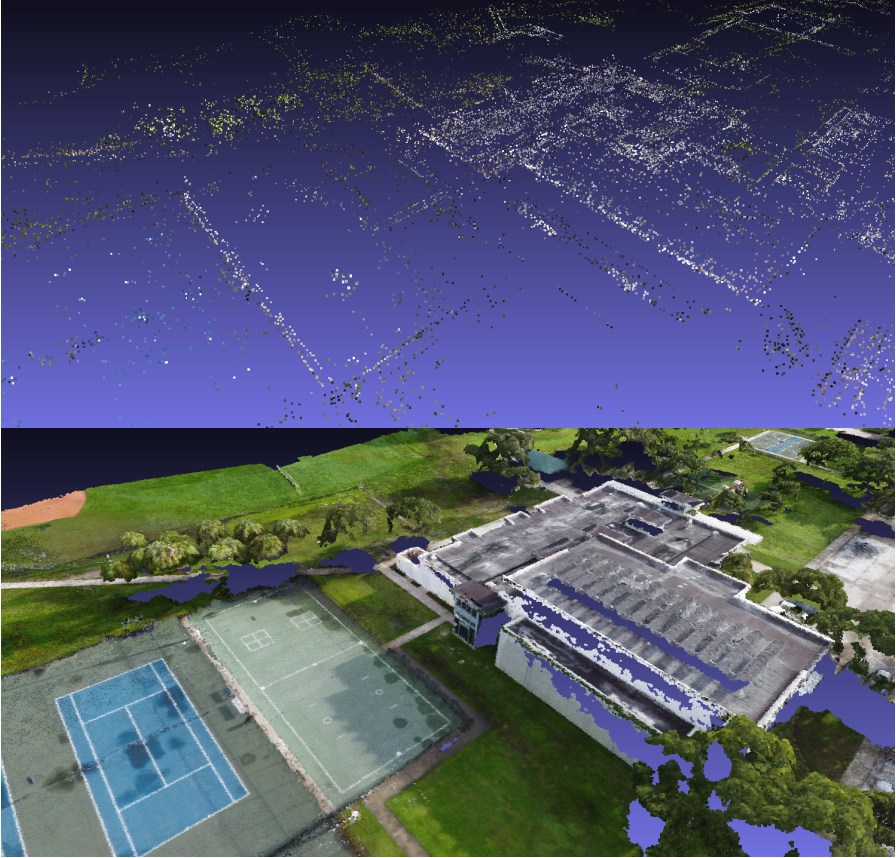
The process of going from point cloud to DEMs is not as straightforward as it may seem. Since DEMs are *rasters* (images), they have *cells* (pixels). Each cell, should have a value. Depending on the resolution of the raster, certain cells may have zero, one or more points that fall into it. Every cell needs a value, even if no points fall directly into it, otherwise there will be empty areas (gaps) in the DEM! One way to overcome this is to use a radius around each cell. Every point that falls within the radius is considered part of the cell.



Pixels and points (left), radius of 0.5 (middle) and radius of 1 (right)

But how big should the radius be? If too small, as in the 0.5 radius example above, some cells might remain empty. If too big, there will be too much smoothing and accuracy will suffer. Since different point clouds have varying degrees of density, one solution is to compute multiple DEMs with different radiuses and stack them.

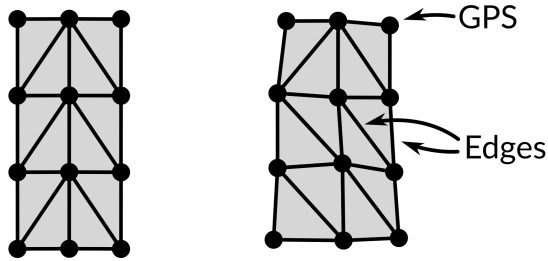
6. Task Options in Depth



Sparse (top) vs. dense (bottom) point cloud outputs

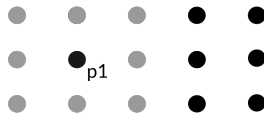
Both point clouds can be used to generate a mesh. However, it's better to have more points, as meshes can be created with more details. In the dense point cloud screenshot above, the building on the right side of the scene is well defined, but it's poorly represented in the sparse point cloud. Buildings are especially difficult to model without a dense point cloud, so this option tends to yield poor results in urban areas. For flat areas such as farmlands, however,

6. Task Options in Depth



Initial graph (left) and graph with randomly moved positions and new edges (right). Every edge indicates an image pair

ODX also supports a different method to perform preemptive matching by considering only the nearest neighbors of each image instead of using a graph. It is enabled by setting this option. The illustration below shows the result of setting this option to 8:



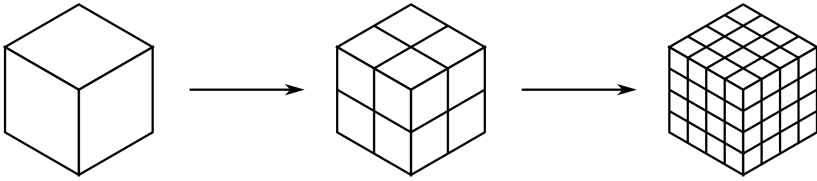
Dots represent approximate image locations, extracted from EXIF tags. When the `matcher-neighbors` is set to 8, only the 8 nearest neighbors (highlighted in gray) are considered for matching with image `p1`

This option can sometimes be beneficial for speeding up processing by reducing the number of matching pairs. If no GPS information is available, this option is disabled and all image pairs are considered, unless `matcher-order` is specified.

6. Task Options in Depth

The details of the algorithm are fascinating, but probably outside the scope of this book.

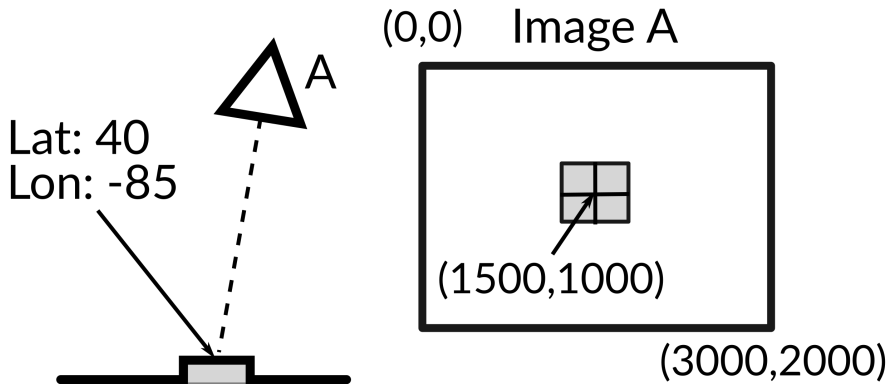
To understand how this option affects the output, it helps to visually understand the concept of an octree. First, octree means *eight-tree* (okta is *eight* in Greek). Why eight? Because at each level (or *depth*) of the tree, each box (or *node or branch*) of the tree is divided in eight parts. At the first level there's only one branch. At the second level there's 8. At the third there's 64 and so forth.



An octree with depth 1, 2 and 3

Lower depths in an octree allow finer details to be captured.

7. Ground Control Points



A GCP marker is photographed by camera A to produce Image A. In a second step, the pixel location of the marker (1500,1000) from Image A can be manually tagged with its real world coordinates (latitude 40, longitude -85)

Using ground control points can increase the georeferencing accuracy of a reconstruction, since measurements of static (non-moving) objects using a high precision GPS are often better than those obtained from the GPS of moving UAVs.

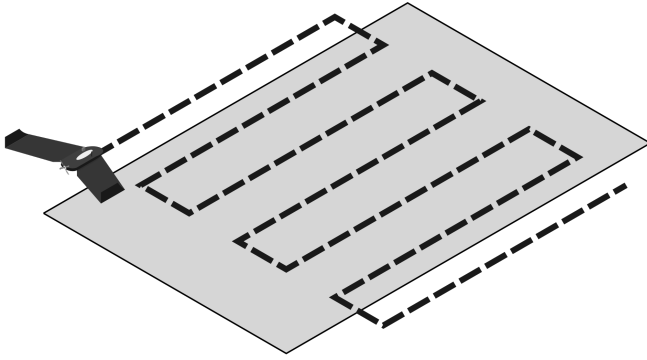
The ideal number of ground control points ranges between 5 to 8, placed evenly across the area to be flown. Adding more than 8 ground control points does not necessarily result in increased accuracy.

If the same marker is visible from multiple images, it should be tagged multiple times for each image. Ideally each marker should be tagged at least 3 times. Another way to think of it is to capture each marker on at least 3 images. This is so that the marker's location can be triangulated during computation.

Ground control points can be used by providing an additional text file along with the input images. The file follows a simple format:

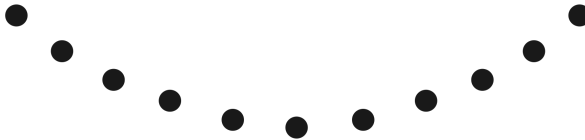
The first line indicates the spatial reference system (SRS) of the world

12. Camera Calibration



Typical flight path from mission planning software. Not great for self-calibration

This doesn't mean a person should never fly this pattern. It just means that when flying this pattern, people need to be aware that the internal camera parameters' estimates will not be as good. Inaccurate parameters lead to an improper camera lens model, which over large areas typically results in a *doming* effect.



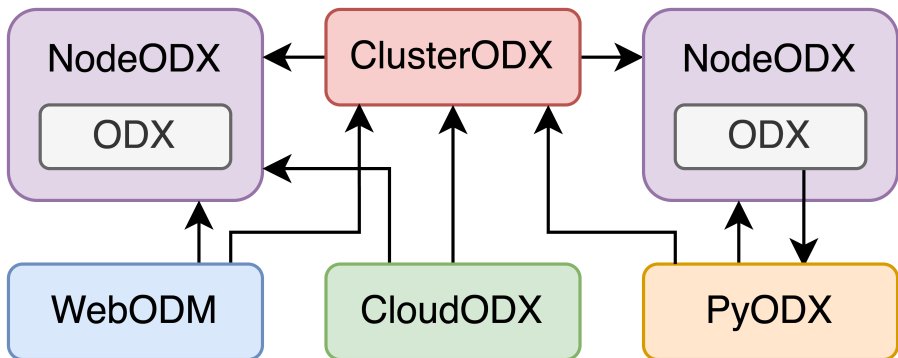
Point cloud exhibiting doming. The terrain appears arched instead of straight

Doming is best cured by following best practices while collecting aerial imagery: flying at different elevations (maximize scale variation) and varying angles.

19. The NodeODX API

ODX is a processing engine and WebODM is a friendly user interface. NodeODX¹ was historically built to allow WebODM to communicate with ODX over a network. Today NodeODX has expanded its role and is the glue that binds together many WebODM projects. Each project understands the API that NodeODX defines. When we say *NodeODX* we are referring to the reference implementation of the NodeODX API available at github.com/WebODM/NodeODX.

At its core, the API defines ways to easily create new tasks, manage such tasks (cancel, delete, restart), download results and query status information.



Many WebODM projects use the NodeODX API to communicate with each other

¹Node is a reference to Node.js, the language NodeODX is written in

Glossary

2.5D Model: A model where elevation is simply *extruded* from the ground plane and thus is not a true 3D model.

Artifacts: undesired alterations generated as the result of a digital process.

API: Application Programming Interface. A set of functions allowing the creation of applications that access the features or data of another application.

Bundle Adjustment: a refinement step during the Structure From Motion process that improves the location of cameras, the 3D points of the scene and the camera parameters.

CloudODX: A command line tool to process aerial imagery in the cloud.

ClusterODX: A NodeODX API compatible autoscalable load balancer and task tracker for connecting multiple NodeODX nodes under a single network address.

CRS: Coordinate Reference System. A CRS is a coordinate-based system used to locate geographical entities.

CSV: Comma Separated Value is a textual file format where fields are typically separated by commas or some other character such as a space or a tab.

cURL: a software providing a library and command-line tool for transferring data using many protocols.

DEM: Digital Elevation Model (either a DSM or a DTM).